Nebula Runtime Overview

Nebula is a distributed inference platform. The runtime separates the control plane from the data plane so that a control-plane upgrade never interrupts live traffic.

A standard deployment runs three control-plane nodes. Each data-plane node keeps a warm model cache and a local request queue, which is why a cold start stays under 800 ms even after a rolling restart.

The runtime retries a failed inference request at most three times. Idempotency keys are recorded for 24 hours so a retried request is never double-charged.

Capacity planning starts from the appliance matrix. The NEBULA-400 appliance sustains 12000 QPS with four accelerators and a 4U footprint.

Operators can drain a node with nebulactl drain --node before maintenance. Drain waits for in-flight requests, then marks the node unschedulable.

Metrics are exported every 15 seconds. The runtime exposes nebula_queue_depth and nebula_tokens_per_second as its primary capacity signals.